F26-12: Can Early Training Dynamics Predict Out-of-Distribution Generalization?

Phase 3 Report: Implementation and Execution

Track A: Research and Development

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Summary. Phase 3 delivers a working end-to-end pipeline on Colored MNIST. We trained 18 small CNNs (6 shortcut strengths x 3 seeds), saved checkpoints at 5, 10, 15, 20 and 100% of training, measured an early colour-reliance signal at each checkpoint, computed the static ATC baseline from the final model, and recorded true accuracy on in-distribution (ID) and two out-of-distribution (OOD) test sets. Every model produces one row of a results table. Final statistical comparisons belong to Phase 4; this report shows that the pipeline works and gives first, honestly limited results.

1. Model and algorithm selection

- **Data (Colored MNIST, generated in Phase 2).** Each digit class has one fixed hue. With probability p an image gets its true class colour, otherwise one of the other nine colours. We use p in {0.10, 0.50, 0.70, 0.90, 0.95, 0.99}; p = 0.10 is the no-shortcut control. Digit shape alone carries the label (logistic regression on grayscale pixels reaches 0.9021 validation accuracy), so a model can succeed without colour. This makes shortcut strength a controlled variable. Splits: 50,000 train, 10,000 validation, 10,000 test.
- **OOD test sets.** *Reversed*: colour never matches the digit (agreement 0.0). *Uncorrelated*: colour is random (agreement 0.1). Both are built from the official MNIST test images.
- **Model: SmallCNN.** Two convolutional blocks (3 to 32 and 32 to 64 channels, each with ReLU and 2x2 max pooling) followed by two linear layers (3136 to 128 to 10). It is small enough to train 18 times, has several layers (needed later for Prediction Depth), and is a standard image classifier.
- **Early signal (our own): feature-ablation reliance.** The relative accuracy drop on the 10,000 ID validation images when the shortcut is changed: (normal accuracy minus changed accuracy) divided by normal accuracy. Two variants: *neutralized* (digit shown in gray, colour removed) and *random* (each image given a new random colour). It uses only ID validation images, no OOD data.
- **Baseline: ATC (Garg et al., ICLR 2022 [1]).** Confidence is the maximum softmax probability. A threshold is chosen on the ID validation set so that the fraction of examples above it equals validation accuracy. The predicted OOD accuracy is the fraction of unlabeled test examples above that threshold.
- **Why only these in Phase 3.** The reliance score and ATC are the simplest to implement and verify. Prediction Depth [2], the gradient-starvation proxy [3] and Agreement-on-the-Line [4] are planned for Phase 4.

2. Implementation details

Python with PyTorch [7] (model, training), NumPy and pandas (data and results table), matplotlib (figures), run in Google Colab on a **CPU** runtime. The Phase 2 objects are reused: make_dataset, the stored colour assignments COLORS, and the channel statistics MEAN and STD. Colours are applied when images are loaded. The code is organised as plain functions: set_seed, evaluate, get_checkpoint_steps, relative_drop, find_atc_threshold, train_one_model (returns one results row) and run_grid (loops over runs, skips finished ones, saves the CSV after every run).

3. Code workflow and pipeline

Stage	What happens
Input	Colored MNIST at shortcut strength p (train and validation), plus ID test, reversed and uncorrelated test sets.
Train	Seed all random generators, train SmallCNN with Adam for a fixed 1,955 steps.
Checkpoints	At steps 98, 196, 294, 391 and 1955 (5, 10, 15, 20, 100%): measure validation accuracy, neutralized accuracy, random-colour accuracy, and save the weights.
Baseline	After training: ATC threshold from the model's ID validation confidences.
Evaluate	True accuracy and ATC prediction on ID test, reversed and uncorrelated sets.
Output	One row per model in phase3_results.csv (signals per checkpoint, ATC prediction, true accuracies), plus summary tables and figures.

4. Training process

Item	Value	Item	Value
Split	50,000 / 10,000 / 10,000	Optimizer	Adam, learning rate 1e-3
Batch size	128 (391 steps per epoch)	Epochs / total steps	5 / 1,955 (fixed in advance)
Loss	Cross-entropy, 10 classes	Normalisation	Per-channel MEAN and STD (Phase 2)
Population	6 values of p x 3 seeds = 18 models	Seeds	0, 1, 2 (set inside every run)
Device	CPU (Google Colab)	Run time	about 6 to 7 min per model, about 2 h in total

Total training length is fixed in advance, so each checkpoint fraction means the same thing for every model. The 20% checkpoint (step 391) is exactly the end of the first epoch. Seeds change the initial weights and batch order; the colour assignments are fixed by the Phase 2 data seed.

5. Initial results

5.1 Sanity check. Table 1 shows the intended behaviour. As p grows, ID test accuracy stays high (0.9886 to 0.9966) while accuracy on the reversed set falls from 0.9884 to 0.7757. The p = 0.10 model is unaffected by colour. The p = 0.99 model degrades clearly but does not collapse to 0 (a pure-colour model would score 0.0 on the reversed set), which is plausible because digit shape also carries the label and about 1% of training images have a mismatched colour.

p	ID test acc	Reversed acc	Reversed std (3 seeds)	Uncorr. acc	ATC error reversed	ATC error uncorr.	Reliance neutral 10%	Reliance random 10%	Reliance neutral 100%
0.10	0.9886	0.9884	0.0017	0.9885	0.0017	0.0020	0.0009	0.0006	-0.0005
0.50	0.9894	0.9842	0.0023	0.9865	0.0023	0.0037	0.0121	0.0107	0.0029
0.70	0.9911	0.9785	0.0019	0.9807	0.0018	0.0013	0.0337	0.0315	0.0068
0.90	0.9923	0.9497	0.0062	0.9550	0.0053	0.0048	0.1261	0.1229	0.0293
0.95	0.9943	0.9331	0.0124	0.9404	0.0080	0.0091	0.1865	0.2306	0.0387
0.99	0.9966	0.7757	0.0087	0.7992	0.0438	0.0340	0.4351	0.6076	0.1197

Table 1. Mean over 3 seeds for each shortcut strength p (18 models). ATC error is the absolute difference between ATC's predicted and the true accuracy. Reliance is the relative accuracy drop defined in Section 1.

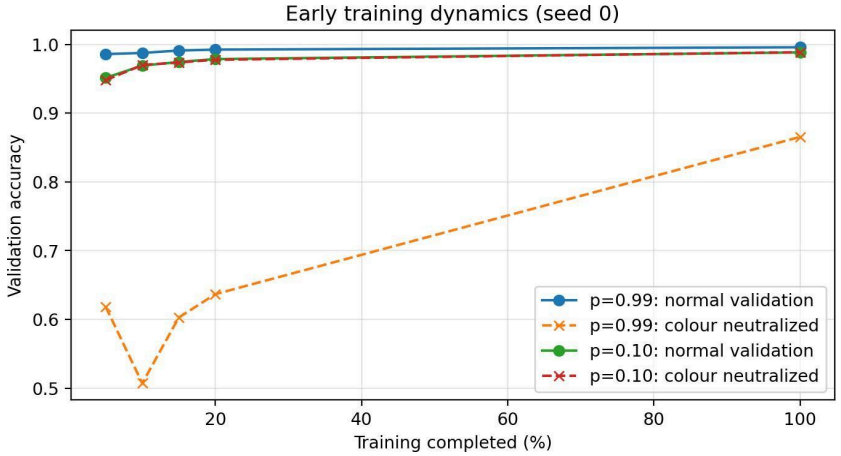


Figure 1. Validation accuracy at the checkpoints (seed 0). Lines between 20% and 100% are straight interpolation because no checkpoint was measured in between.

5.2 Early signal. In Figure 1 the p = 0.99 model already reaches 0.986 normal validation accuracy at 5% of training, yet its colour-neutralized accuracy is only 0.618, 0.507, 0.603 and 0.637 at 5 to 20%, recovering to 0.865 at the end. The p = 0.10 model shows no gap. Mean reliance at p = 0.99 falls from 0.4351 (neutralized) and 0.6076 (random) at 10% to 0.1197 (neutralized) at 100%, so reliance on colour is largest early. Random-colour reliance is larger than neutralized reliance at high p, because a misleading colour hurts more than a missing colour.

5.3 ATC baseline. ATC is accurate here. Its mean absolute error is below 1 point for p up to 0.95 on both OOD sets and 4.4 (reversed) and 3.4 (uncorrelated) points at p = 0.99. For the seed 0 model at p = 0.99 it overestimates (predicted 0.8298 against true 0.7658 on the reversed set).

5.4 Do early signals predict OOD accuracy? Across all 18 models the correlation between early reliance (10%) and true reversed accuracy is -0.95 (neutralized) and -0.98 (random); ATC's prediction correlates +0.97. These pooled values are driven by p, which is known before training. After subtracting each p's mean, the correlations with true reversed accuracy are +0.18 (random reliance), -0.05 (neutralized reliance) and +0.23 (ATC), so no signal shows a detectable within-p relation (3 seeds per p is a very small sample). As a reference, a per-p average lookup has an in-sample mean absolute error of 0.0040 against 0.0105 for ATC. We therefore treat the pooled correlations as descriptive only. They do not yet support or reject H1 to H4; testing within-p and held-out-p prediction is the work of Phase 4.

6. Challenges faced

- **Sanity check was stricter than reality.** We expected the p = 0.99 model to collapse on the reversed set; it fell to 0.78 instead. We redefined the check as clearly worse than p = 0.10 and clearly below its own ID accuracy, which holds (0.7757 vs 0.9884).
- **Ablation confound.** The neutralized input (a gray digit normalised with coloured-image statistics) is an input the high-p model never saw, so part of its drop may come from unfamiliar input. We added the random-colour variant, which stays inside the training colour range. The random variant is close to the uncorrelated OOD test by construction, which must be kept in mind when interpreting correlations.
- **First run was not reproducible and not reusable.** The first two-model run set the seed only once and saved only the final weights. We now seed inside every run and save the weights at all five checkpoints (needed for Prediction Depth and the gradient proxy).
- **Pooled correlations are misleading.** The strong overall correlations disappear within each p (Section 5.4).
- **Run time.** About 2 hours for 18 CPU runs. We made the grid resumable (finished runs are skipped, the CSV is saved after every run). The images are tinted on the fly; precomputing tinted tensors and using a GPU is planned for Phase 4.
- **Short early window.** The first 20% of a 5-epoch budget is a single epoch, and all models are already near 98% ID accuracy there.

7. Limitations and next steps (Phase 4)

Not yet done: Prediction Depth, the gradient-starvation proxy (definition still open), Agreement-on-the-Line, Waterbirds and a second architecture. ATC and the early signals use different information: ATC sees unlabeled OOD inputs, while the reliance signals use only ID validation images. Phase 4 will report this openly and add a variant of the early signals that uses unlabeled OOD data. Planned: more seeds at p of 0.90 and above (where variation exists), more sources of model diversity (learning rate, width, epochs, architecture), a "p only" reference in every comparison, regression and correlation metrics (Pearson, Spearman, MAE, RMSE, R-squared), cross-validation with held-out p, and bootstrap confidence intervals.

References

[1] S. Garg et al. Leveraging Unlabeled Data to Predict Out-of-Distribution Performance. ICLR 2022 (ATC).

[2] R. Baldock, H. Maennel, B. Neyshabur. Deep Learning Through the Lens of Example Difficulty. NeurIPS 2021 (Prediction Depth).

[3] M. Pezeshki et al. Gradient Starvation: A Learning Proclivity in Neural Networks. NeurIPS 2021.

[4] C. Baek et al. Agreement-on-the-Line: Predicting the Performance of Neural Networks under Distribution Shift. NeurIPS 2022.

[5] M. Arjovsky et al. Invariant Risk Minimization. 2019 (Colored MNIST idea).

[6] Y. LeCun, C. Cortes, C. Burges. The MNIST database of handwritten digits.

[7] A. Paszke et al. PyTorch: An Imperative Style, High-Performance Deep Learning Library. NeurIPS 2019.